

Blockchain-based Digital Money for Bank Indonesia Business Process

1st Muhammad Fachrizal Giffari

Department of Electrical and
Information Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia

muhammadfachrizalgiffari@mail.ugm.ac.id

2nd Teguh Bharata Adji

Department of Electrical and
Information Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia

adji@ugm.ac.id

3rd Guntur Dharma Putra

Department Electrical and
Information Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia

gdputra@ugm.ac.id

Abstract—Central Bank Digital Currency (CBDC) is a strategic Bank Indonesia initiative under Project Garuda, but the publicly documented Garuda experiments focus on the wholesale tier, leaving a technical gap for a testable system that digitizes Bank Indonesia’s retail money business processes (issuance, two-tier distribution, risk-based customer onboarding, payment, redemption, and supervision) under one auditable blockchain ledger. This paper designs and implements such a system: a retail Digital Rupiah prototype on a permissioned five-organization Hyperledger Fabric network. Go chaincode stores only KYC status, risk level, due-diligence level, senior-approval status, expiry, and document hashes, while raw identity data remains in an off-chain PostgreSQL store. Retail wallet tiers (BASIC, STANDARD, MERCHANT) are derived deterministically from the approved KYC profile rather than selected by the user, and each tier enforces per-transaction, daily-outgoing, monthly-outgoing, monthly-incoming, and maximum-balance limits computed from the Fabric transaction timestamp for endorsement determinism. Institutional Bank Indonesia, bank, and payment-service-provider wallets remain outside retail tiering and use system-level limits. We formalize the transfer-validation predicate, the tier-derivation function, and the double-spend-safety argument, and evaluate performance with a reproducible Hyperledger Caliper protocol that sweeps client concurrency and exercises the business processes directly. Across transfer rounds at three concurrency levels and four business-process workloads the prototype processed 616 transactions with zero failures, zero MVCC read conflicts, and zero policy or limit rejections. Transfer throughput saturates near 14.5 TPS independent of client concurrency while latency stays bounded (mean ≤ 0.81 s, maximum ≤ 2.32 s), and monetary operations reach 19.9 TPS at 0.32 s mean latency. The result is an auditable technical basis for studying how Bank Indonesia’s business processes can be carried on blockchain-based digital money without claiming production readiness.

Index Terms—Digital Rupiah, retail CBDC, Hyperledger Fabric, risk-based KYC, wallet limits, business process, blockchain benchmarking

I. INTRODUCTION

Central banks worldwide are evaluating Central Bank Digital Currency (CBDC) as a digital form of sovereign money [1]. In Indonesia, Bank Indonesia pursues this agenda through Project Garuda, whose publicly documented proof of concept targets the *wholesale* tier: a cash ledger for issuance and redemption among the central bank and large financial institutions [2], [3]. The retail tier, the distribution of Digital Rupiah to households and merchants through intermediary banks and

payment service providers (PJP), is described at the level of policy and architecture in the Garuda White Paper [3], but no openly available, testable system carries Bank Indonesia’s retail money business processes end to end on a blockchain ledger.

Digital money for a central bank is not a single ledger operation but a chain of business processes: the central bank *issues* money against reserves, banks *distribute* liquidity downstream to PJP, PJP *onboard* customers and merchants under risk-based Know-Your-Customer (KYC) checks, holders *transact* within category-specific limits, intermediaries *redeem* money back to the central bank, and the supervisor *reconciles* and audits the whole circulation. A blockchain-based realization must encode each of these processes as deterministic, independently validated transactions while preserving two-tier intermediation that protects banks from disintermediation [4] and exposing an audit surface to the supervisor. Prior technical work tends to isolate single concerns: permissioned CBDC architecture [5], privacy designs [6], security threat models [7], or raw throughput on account-based systems [8]. Deployed retail systems such as Nigeria’s eNaira [9], China’s e-CNY [10], and the Bahamian Sand Dollar [11] publish design papers but not reproducible performance benchmarks, and the surveyed literature lacks an integrated, openly measurable retail reference for the Indonesian two-tier model [12], [13].

This paper addresses that gap with a Digital Rupiah prototype on a permissioned Hyperledger Fabric [14] network that maps Bank Indonesia’s business processes onto chaincode transactions. The contributions are:

- 1) An integrated two-tier Digital Rupiah chaincode that unifies hierarchical issuance and distribution, risk-based KYC with deterministic wallet-tier derivation, tiered transfer limits, and supervisor-facing reconciliation in a single artifact, completing the retail design beyond the wholesale-only Garuda proof of concept.
- 2) A formal, deterministic, double-spend-safe transfer protocol that combines a tier-limit validity predicate, Fabric multi-version concurrency control (MVCC), and a transaction-timestamp clock, so that limit counters and validation are identical across endorsing organizations.
- 3) A reproducible open retail-CBDC benchmark with Hy-

TABLE I
COMPARISON WITH REPRESENTATIVE RETAIL CBDC SYSTEMS

System	Platform	2-tier	KYC+lim.	Bench.
e-CNY [10]	Permissioned	✓	n/r	×
eNaira [9]	Fabric	✓	n/r	×
Sand Dollar [11]	Permissioned	✓	n/r	×
OpenCBDC [8]	Account (non-DLT)	×	×	✓
This work	Fabric 2.5	✓	✓	✓

n/r: not publicly reported. Bench.: open, reproducible throughput/latency benchmark.

perledger Caliper [15] that documents workload, population seeding, a client-concurrency sweep, and business-process profiles, and that locates the throughput saturation point and verifies the business processes under load.

II. BACKGROUND AND RELATED WORK

A. Two-Tier Retail CBDC

In the two-tier (intermediated) model the central bank issues CBDC and remains the ultimate liability holder, while regulated intermediaries onboard end users, perform KYC, and operate customer-facing rails [1], [4]. This preserves the existing division of labor and limits the central bank’s exposure to retail operations. Bank Indonesia adopts this structure for Digital Rupiah, with banks and PJP acting as distributors between the central bank and the public [3].

B. Permissioned Blockchain and Hyperledger Fabric

Hyperledger Fabric is a permissioned, modular blockchain with pluggable ordering, channel-based data isolation, and an execute–order–validate transaction flow [14]. Identity is governed by Membership Service Providers (MSPs); ordering uses the crash-fault-tolerant Raft protocol; and the world state can be backed by CouchDB to support rich queries. These properties make Fabric a common substrate for CBDC and regulated-finance prototypes [5], [16]. Standard frameworks such as Hyperledger Caliper [15], [17] provide measurement protocols for throughput and latency under varying load.

C. Compliance, Limits, and Supervision

Retail CBDC policy work stresses embedding AML/CFT controls and holding limits into the design rather than bolting them on, which in Indonesia is governed by anti-money-laundering regulation [18]. Token-based retail designs are also exposed to denial-of-service and double-spend risks [19], motivating deterministic on-chain enforcement. This work integrates these concerns (tiered limits, KYC gating, and supervision events) into one chaincode, complementing single-aspect studies of privacy [6] and security [7]. Table I positions the prototype against representative deployed and prototype retail CBDC systems: it is the only entry that combines two-tier intermediation, an integrated KYC-and-limit policy, and an openly reproducible benchmark.

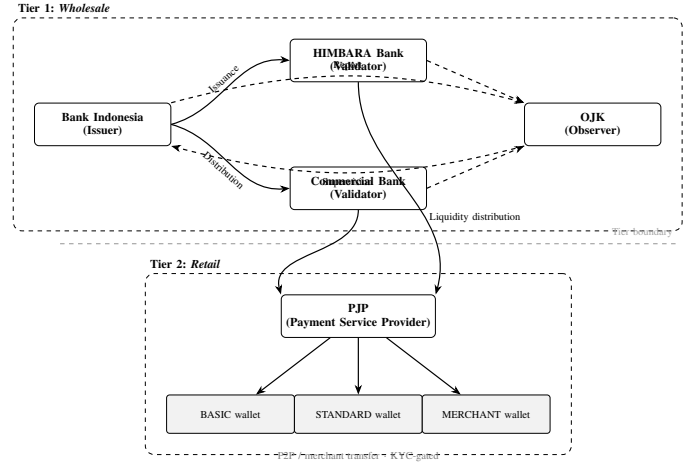


Fig. 1. Two-tier actor structure of the Digital Rupiah system. Bank Indonesia issues and distributes to validator banks (Tier 1); banks distribute liquidity to the PJP, which serves tiered retail wallets (Tier 2). The OJK observer receives reports and supervises.

TABLE II
BANK INDONESIA BUSINESS PROCESSES AND THEIR CHAINCODE REALIZATION

Process	Initiator	Chaincode transaction
Issuance	Bank Indonesia	Mint, RequestIssuance
Distribution	Validator bank	DistributeToParticipant
Onboarding/KYC	Bank / PJP	SubmitKycProfile, RefreshKycProfile, CreateWallet
Payment	Customer	Transfer
Redemption	Participant	Burn, RequestRedemption
Supervision	OJK / BI	GetReconciliationReport, GetMetrics, GetSupervisionEvents

All state-changing transactions commit under a MAJORITY endorsement policy (3 of 5 organizations).

III. BANK INDONESIA BUSINESS PROCESSES

The system digitizes six business processes that together describe the circulation of central-bank digital money. Fig. 1 shows the two-tier actor structure, and Table II maps each process to its initiating actor, the chaincode transaction that realizes it, and the endorsement it requires.

The wholesale tier comprises institutional participants (Bank Indonesia, banks, and PJP) registered through a participant lifecycle (submit, approve, freeze, unfreeze, reject, offboard). Bank Indonesia mints Digital Rupiah to a treasury wallet; banks request issuance and distribute liquidity downstream, with a chaincode guard that permits validator-to-PJP distribution but blocks a PJP from requesting direct issuance. The retail tier comprises customers and merchants served by the PJP. Institutional wallets are not retail-tiered; they are bounded by system-level limits.

IV. SYSTEM ARCHITECTURE

The architecture follows a four-layer stack (client, backend API, off-chain store, and Fabric network) shown in Fig. 2. A web client calls a Go REST backend (Gorilla Mux) that

TABLE III
NETWORK TOPOLOGY OF THE FIVE-ORGANIZATION FABRIC NETWORK

Organization	Role	Peer	CouchDB
BankIndonesiaOrg	Issuer	peer0.bi:7051	:5984
HimbaraBankOrg	Validator	peer0.himbara:9051	:6984
CommercialBankOrg	Validator	peer0.commercial:11051	:7984
PJPOrg	Distributor	peer0.pjp:13051	:9984
OJKObserverOrg	Observer (R/O)	peer0.ojk:12051	:8984
Ordering: orderer.paynet:7050 (Raft CFT); per-org CA; TLS enabled			

authenticates requests with JWT and role-based access control, persists off-chain KYC and authentication data in PostgreSQL, and submits transactions to the ledger through the Fabric Gateway SDK. The permissioned network itself is detailed in Fig. 3.

A. Network Topology

The network (Fig. 3, Table III) runs five peer organizations on one application channel: *BankIndonesiaOrg* (issuer), *HimbaraBankOrg* and *CommercialBankOrg* (validator/distributor banks), *PJPOrg* (payment service provider), and *OJKObserverOrg* (a read-only supervisor). Each organization runs one peer with a CouchDB world state and its own certificate authority; identity is governed by per-organization MSPs; and ordering uses a single Raft orderer. The chaincode endorsement policy requires a MAJORITY of organizations to endorse before a transaction commits. With five organizations in the network, that means at least three peers (a simple majority) must independently validate a transaction before it is accepted.

B. On-Chain and Off-Chain Data

The chaincode (`digital_rupiah.go`, Go) groups functions into participant management, wallet management, issuance and redemption, retail transfer, KYC and compliance, limit management, and supervision (Table IV). The on-chain KYC anchor stores only status, subject type, risk level, due-diligence level, senior-approval flag, expiry, and document hashes. All personally identifiable identity data resides off-chain in PostgreSQL, whose schema (Fig. 4) centers on a `kyc_profiles` hub linked to provider-check and audit-event tables, with an isolated `auth_users` table for portal authentication.

A Go backend enforces role-based access control over seven roles, summarized in Table V, before any transaction reaches the chaincode.

V. PROTOCOL AND DETERMINISM

This section formalizes the compliance logic carried by the chaincode and argues its safety under concurrent endorsement.

TABLE IV
CHAINCODE FUNCTION GROUPS

Group	Representative functions
Participants	SubmitParticipant, ApproveParticipant, Freeze/Unfreeze, Offboard
Wallets	CreateWallet, CreateWholesaleWallet, Freeze/UnfreezeWallet
Issuance	Mint, RequestIssuance, RequestIssuanceRtgs
Redemption	Burn, RequestRedemption
Distribution	DistributeToParticipant
Transfer	Transfer
KYC	SubmitKycProfile, RefreshKycProfile, GetKycProfile
Limits	SetTierLimit, SetSystemLimit
Supervision	GetReconciliationReport, GetMetrics, GetSupervisionEvents

TABLE V
BACKEND ROLES AND REPRESENTATIVE PERMISSIONS

Role	Representative permitted actions
Public	Health check, login
Authenticated	View own profile, list wallets
KYC-verified	Initiate transfer, view balance, redeem
Bank/PJP	Onboard customer, submit KYC, create wallet, distribute
Bank Indonesia	Issue, set system limits, approve participant
Merchant	Initiate transfer, view balance
Supervisor	Reconciliation, metrics, supervision events

A. Wallet-Tier Derivation

Let s be the subject type, d the due-diligence level, and r the risk level of an approved KYC profile. The wallet tier τ is a deterministic function

$$\tau(s, d, r) = \begin{cases} \text{BASIC} & d=0, s=C, r=0 \\ \text{STANDARD} & d \geq 1, s=C, r \leq 1 \\ \text{MERCHANT} & d \geq 1, s=Mr, r \leq 1 \\ \perp & \text{otherwise,} \end{cases} \quad (1)$$

where the subject $s \in \{C \text{ (customer), } Mr \text{ (merchant)}\}$, the due-diligence level $d \in \{0, 1, 2\}$ encodes simplified/standard/enhanced, the risk level $r \in \{0, 1, 2\}$ encodes low/medium/high, and $\perp$ denotes a profile that cannot back a retail wallet. A high-risk subject additionally requires enhanced due diligence and senior approval, and a prohibited risk level is rejected. Because τ depends only on the approved profile, the tier is derived rather than user-selected.

B. Transfer-Validity Rules

Each wallet tier has five limits, listed in Table VI: a per-transaction cap, a daily outgoing cap, a monthly outgoing cap, a monthly incoming cap, and a maximum balance. For a transfer of amount a from a sender to a receiver, the chaincode admits the transfer only if every one of the following checks passes:

- neither the sender's nor the receiver's wallet is frozen;
- the sender's KYC profile is approved and not expired;

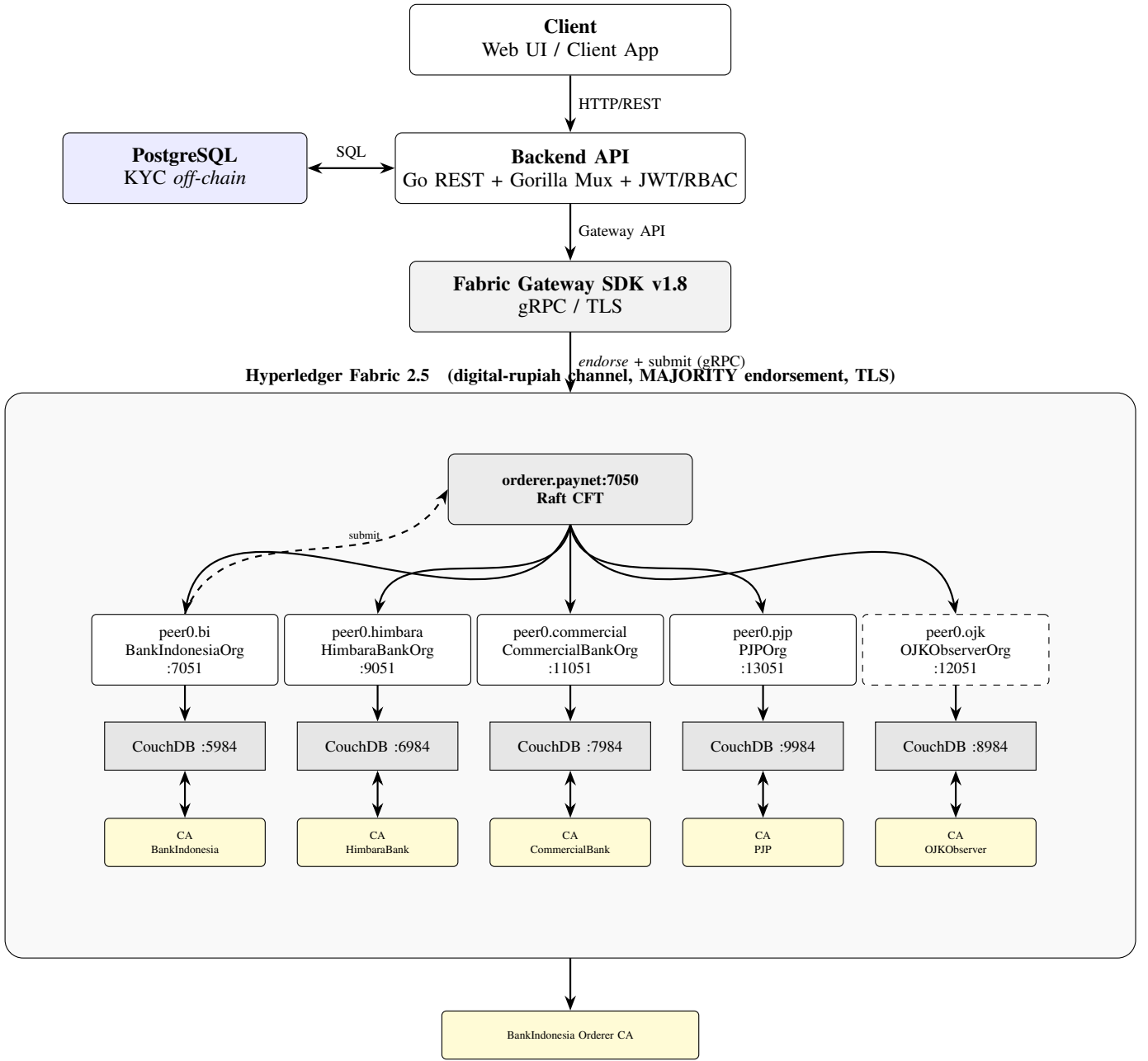


Fig. 2. Full system architecture of the Digital Rupiah prototype: client layer, Go REST backend with JWT/RBAC, off-chain PostgreSQL store, Fabric Gateway SDK, and the Hyperledger Fabric 2.5 network of five peer organizations with CouchDB, a per-organization CA, and a Raft orderer.

- the amount does not exceed the sender's per-transaction cap;
- the sender's running daily and monthly outgoing totals, plus this transfer, stay within the sender's daily and monthly outgoing caps;
- the receiver's running monthly incoming total, plus this transfer, stays within the receiver's monthly incoming cap;
- the sender has enough balance to cover the transfer; and
- the resulting receiver balance does not exceed the receiver's maximum-balance cap.

If any check fails, the transfer is rejected. Institutional wallets (Bank Indonesia, banks, and PJP) skip these tier limits but are still bounded by the system-level caps in Table VII: a total supply cap on the sum of all balances, a per-participant balance cap, and a per-transaction cap.

C. Determinism and Double-Spend Safety

Because every endorsing peer evaluates the rules above independently, the daily/monthly counters and the KYC-expiry check must not depend on each peer's own wall clock, or different peers could reach different answers. To avoid this, the chaincode reads Fabric's transaction timestamp, which is

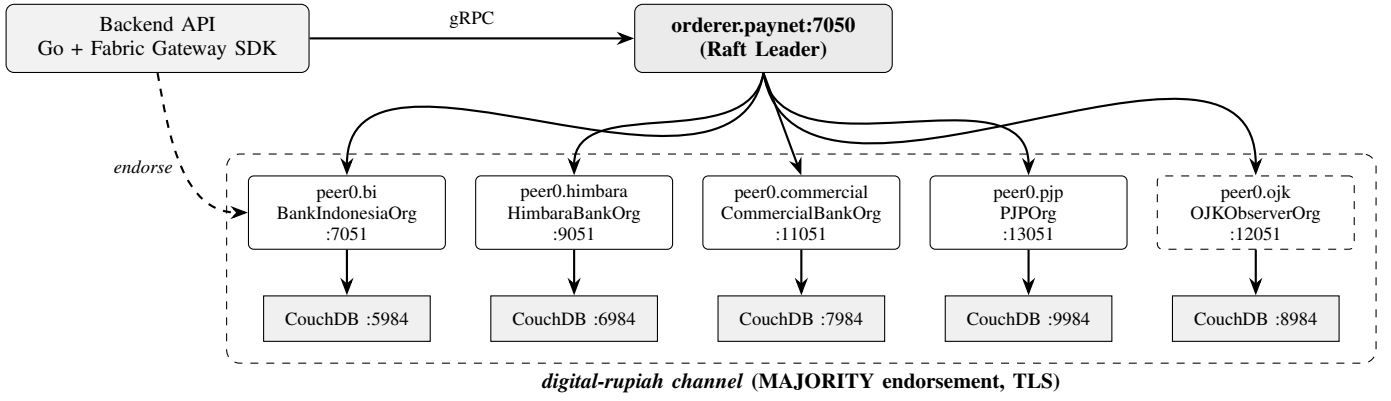


Fig. 3. Hyperledger Fabric network topology: five peer organizations, each with a CouchDB world state, on a single channel under a Raft orderer; the Go backend endorses via the Fabric Gateway SDK.

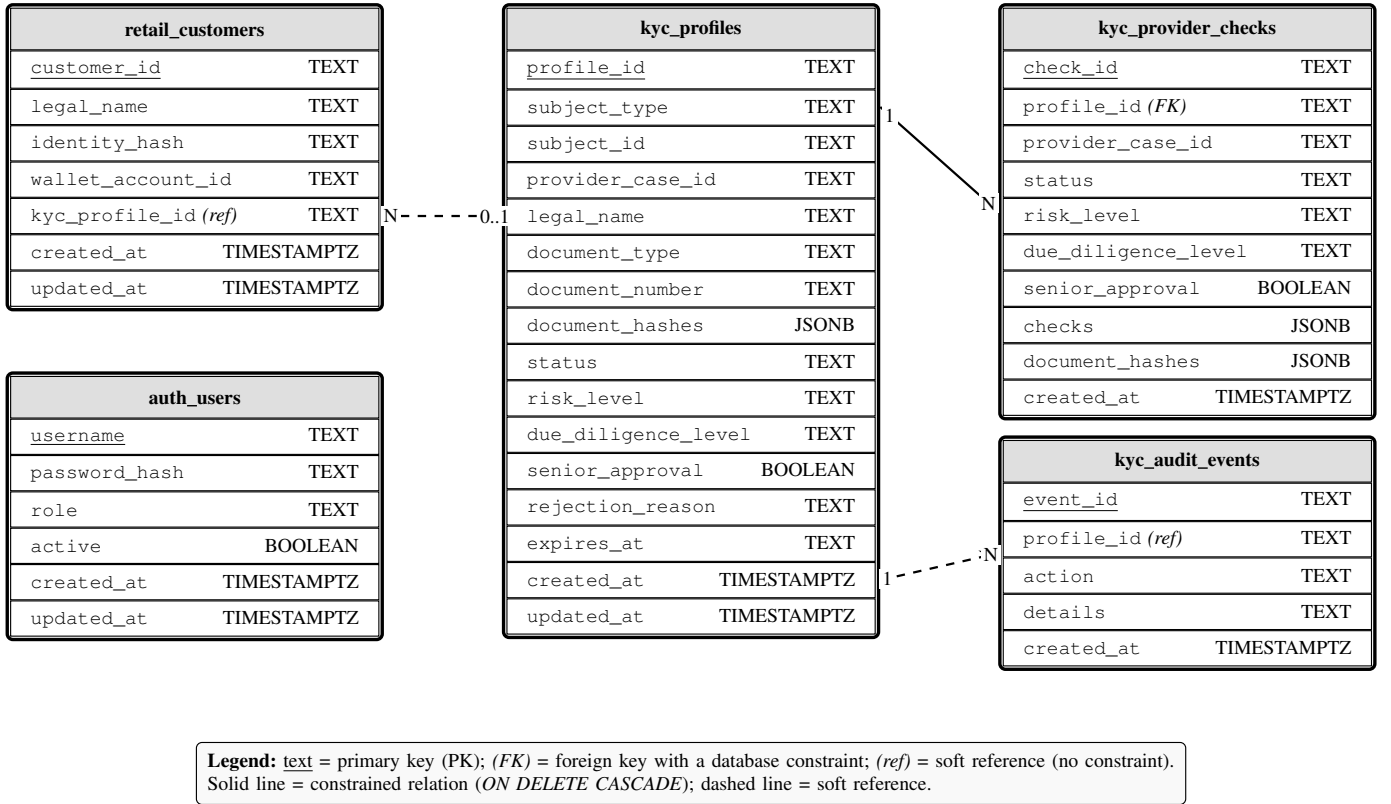


Fig. 4. Off-chain PostgreSQL entity-relationship schema. The auth_users table stands alone for portal authentication, while kyc_profiles is the hub linking the provider-check history and the audit trail. No raw identity data is written to the ledger.

TABLE VI
RETAIL WALLET TIER LIMITS (INDONESIAN RUPIAH)

Limit	BASIC	STANDARD	MERCHANT
Max. balance	2,000,000	20,000,000	200,000,000
Per transaction	250,000	2,500,000	10,000,000
Daily outgoing	500,000	10,000,000	50,000,000
Monthly outgoing	5,000,000	40,000,000	500,000,000
Monthly incoming	20,000,000	40,000,000	500,000,000

TABLE VII
SYSTEM-LEVEL LIMITS (INSTITUTIONAL WALLETS)

Limit	Meaning	Value (Rp)
Total supply cap	Sum of all balances	10,000,000,000,000
Per-participant cap	Per-participant balance	1,000,000,000,000
Per-transaction cap	Per-transaction amount	100,000,000,000

identical for every endorser of a given transaction, and uses

```

1: function TRANSFER( $u, v, a$ )
2:  $\zeta \leftarrow \text{GETTXTIMESTAMP}()$ 
3: if frozen( $u$ )  $\vee$  frozen( $v$ )  $\vee \neg \text{kycOK}(u, \zeta)$  then
4:   return REJECT
5: end if
6: compute  $o_d(u), o_m(u), i_m(v)$  from history using  $\zeta$ 
   {daily/monthly totals, see Sec. V}
7: if  $\neg \text{VALID}(u, v, a)$  then
8:   return REJECT
9: end if
10:  $b(u) \leftarrow b(u) - a$ ;  $b(v) \leftarrow b(v) + a$ 
11: append record to  $H(u), H(v)$ ; emit supervision event if
   high-risk
12: PutState (validated by Fabric MVCC)
13: return COMMIT

```

Fig. 5. Deterministic, double-spend-safe transfer executed on every endorsing peer.

it to decide which day or month window a transfer falls into. Each wallet’s daily and monthly outgoing and incoming totals are simply the sum of its prior transfers that fall inside that window, so every endorser computes exactly the same totals and reaches the same accept-or-reject decision.

Double spending is prevented by combining this deterministic check with Fabric’s built-in multi-version concurrency control (MVCC): every endorsement reads the current version number of a wallet’s balance, and Fabric only commits a transaction if that version number has not changed by the time the transaction reaches the ordering and validation stage. If two transfers try to spend from the same wallet at the same time, both endorse against the same starting balance and version, but only the first one to be ordered commits; the second fails MVCC validation and is rejected, even though it independently passed the balance check. Algorithm 1 states the transfer routine.

VI. PERFORMANCE EVALUATION

A. Methodology

Performance is measured with Hyperledger Caliper on Fabric 2.5.16, consistent with blockchain performance-metric guidance [17]. The ledger is seeded with the population of Table IX: 200 retail customers (20% BASIC, 80% STANDARD) and 50 merchants, all funded within their tier caps and holding approved KYC anchors before measurement. Two workload families are run. The first is a transfer workload (70% customer-to-customer, 25% customer-to-merchant, 5% reads; Table VIII) executed at three client-concurrency levels (1, 2, and 4 Caliper workers), each with warmup, sustained, and peak rounds, to probe scaling. The second is a set of four functionality profiles that exercise the business processes of Section III: KYC onboarding, monetary operations (issuance, distribution, redemption), supervision reads, and admin/policy. For each round we report two numbers: throughput, the number of successful transactions divided by the round’s wall-clock duration, and efficiency η , the throughput divided by

TABLE VIII
CALIPER TRANSFER WORKLOAD COMPOSITION

Share	Operation	Transaction
70%	Customer-to-customer transfer	Transfer
25%	Customer-to-merchant transfer	Transfer
5%	Wallet read	GetWallet

TABLE IX
SEEDED TEST POPULATION AND WORKLOADS

Parameter	Value
Retail customers	200 (20% BASIC, 80% STANDARD)
Merchants	50 (MERCHANT tier)
Client concurrency	1, 2, 4 Caliper workers
Transfer rounds	warmup / sustained / peak
Functionality profiles	KYC, monetary, supervision, admin
Total transactions	616

TABLE X
TRANSFER THROUGHPUT SCALING BY CLIENT CONCURRENCY

Wk.	Round	Send (TPS)	Thr. (TPS)	η	Avg (s)	Max (s)
1	Warmup	10.3	7.8	0.76	0.78	2.14
1	Sustained	30.8	12.7	0.41	0.62	2.15
1	Peak	57.6	14.5	0.25	0.66	2.32
2	Warmup	10.5	7.8	0.74	0.80	2.14
2	Sustained	31.6	12.3	0.39	0.64	2.25
2	Peak	61.0	14.5	0.24	0.69	2.27
4	Warmup	11.1	8.0	0.72	0.81	2.18
4	Sustained	33.3	12.9	0.39	0.64	2.16
4	Peak	65.9	14.5	0.22	0.70	2.30

All rounds: 40 transactions, 0 failures. Wk.: Caliper workers.

the offered (requested) transaction rate. An efficiency near 1 means the system keeps up with the offered load; an efficiency well below 1 means the system is saturated and falling behind. Reported figures derive from the Caliper report generated against the deployed chaincode on a clean ledger.

B. Results

The run committed 616 transactions with zero failures (100% success), zero MVCC read conflicts, and zero policy or limit rejections across all workloads, so the deterministic predicate of Section V held under concurrent endorsement. Table X reports the transfer scaling. Throughput saturates near 14.5 TPS in the peak round and, critically, does not increase as client concurrency rises from one to four workers, indicating a server-side bottleneck (endorsement, ordering, and CouchDB commit) rather than a client limit. Efficiency therefore falls from $\eta \approx 0.75$ at warmup to $\eta \approx 0.24$ at peak, locating the saturation knee below the sustained offered load. Latency nonetheless stays bounded across all rounds: mean at most 0.81 s and maximum at most 2.32 s.

Table XI reports the four business-process profiles. Monetary operations achieve the highest throughput (19.9 TPS) at the lowest latency (0.32 s mean), as they are short single-key writes. Supervision reads sustain 17.0 TPS but at higher

TABLE XI
BUSINESS-PROCESS FUNCTIONALITY PROFILES

Profile	Tx	Send (TPS)	Thr. (TPS)	Avg (s)	Max (s)
KYC onboarding	56	4.4	3.7	0.88	2.29
Monetary ops	80	20.5	19.9	0.32	0.53
Supervision reads	80	30.8	17.0	1.10	3.24
Admin / policy	40	5.3	5.2	0.91	1.72

latency (1.10 s mean, 3.24 s maximum) because they issue CouchDB rich queries. KYC onboarding (3.7 TPS) and admin/policy (5.2 TPS) are multi-step and lower-volume but complete without failure, confirming the business processes operate correctly on the ledger.

C. Discussion

The measured region is appropriate for a single-host functional prototype rather than national capacity. Transfer throughput plateaus near 14.5 TPS and monetary operations near 20 TPS; both sit below the order-of-magnitude wholesale traffic threshold (around 30 TPS) reported for permissioned DLT, consistent with private-DLT retail prototypes that operate far below centralized card networks. That throughput does not rise from one to four client workers isolates the bottleneck to the server side, so future scaling work should target endorsement parallelism, block-cutting parameters, and CouchDB indexing rather than client concurrency. The zero MVCC conflicts and zero limit rejections across 616 transactions provide empirical evidence that the determinism and double-spend arguments of Section V hold under load. Cross-platform throughput comparisons are not directly meaningful: differences in platform, measurement tool, network size, and workload scope make absolute TPS values incomparable across studies. The value here is a documented, reproducible protocol (fixed workload mix, seeded population, concurrency sweep, and business-process profiles) under which the saturation point can be re-measured.

VII. LIMITATIONS AND FUTURE WORK

The evaluation runs on a single host with one peer per organization and no real geographic distribution or network latency, so the Caliper figures apply to the reported configuration and do not generalize to national scale. The KYC process is simulated programmatically rather than integrated with a real eKYC provider. The ledger offers no transaction privacy beyond authorized-peer visibility; no private data collections or privacy-preserving cryptography are used. The design implements neither a payment-queue gridlock-resolution mechanism nor offline, network-disconnected payment, the latter requiring secure devices and post-reconnection settlement as studied in Project Polaris [20]. External integration with domestic financial infrastructure (for example BI-RTGS and BI-FAST) is stubbed rather than implemented, so the prototype satisfies internal integration but not the full interoperability and interconnection mandated for Digital Rupiah. Future work should

re-run the benchmark across distributed multi-host topologies, integrate real eKYC and identity controls, evaluate privacy-preserving supervision, and extend the rails toward gridlock resolution and offline payment.

VIII. CONCLUSION

This paper presented a blockchain-based realization of Bank Indonesia’s digital money business processes: a two-tier Digital Rupiah prototype on a five-organization Hyperledger Fabric network that integrates hierarchical issuance and distribution, risk-based KYC with deterministic wallet-tier derivation, tiered transfer limits enforced with a transaction-timestamp clock, and supervisor-facing reconciliation in a single chain-code. We formalized the tier-derivation function, the transfer-validity predicate, and the double-spend-safety argument, and a reproducible Hyperledger Caliper protocol processed 616 transactions with zero failures, zero MVCC conflicts, and zero policy or limit rejections across a client-concurrency sweep and four business-process workloads, with transfer throughput saturating near 14.5 TPS at bounded latency. By closing the wholesale-only gap left by the public Project Garuda experiments with an auditable, openly measurable retail reference, the prototype offers a technical basis for studying how central-bank business processes can be carried on blockchain-based digital money without claiming production readiness.

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